

BIOL 414 Principles of Biology 2 LABORATORY, Spring 2026

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Office/Phone: P559, 603-641-4129
Office Hours: Tuesday 1-2pm or by appointment
Class meets: Tuesday, 9:10am in Room 380 for an All-BIOL 414 Diversity of Life session, then will proceed to room 554 for lab

Laboratory Objectives:

This lab supports the content covered in the BIOL414 lecture. In the lecture, you will be introduced to the domains and kingdoms of life as well as the physiology of cells, tissues, organs, and organ systems, evolution; and human impact on the biosphere. You should have been provided with specific course objectives for the lecture section. In the lab, we will reinforce aspects of this material and begin to develop skills that will be required for upper-level lab experiences. Over the course of the semester, it is my aspiration to have you:

- Have hands-on experiences with specimen, both living and preserved, that represent various types of life.
- Develop an understanding of phylogeny (the relatedness of different branches of life)
- Demonstrate a basic level of proficiency with common lab equipment and techniques, including weighing & measuring, pipetting, safe use of heat sources, aseptic technique, light microscopy and more.
- Design experiments including development of hypotheses and integration of adequate controls.
- Interpret and analyze scientific data
- Communicate your results in the form of graphs, presentations and posters in a way that can be clearly understood by your audience.
- And enjoy the experience!*

Lab Manual:

You'll be provided the lab handouts you need to complete our lab activities, or they will be posted to Canvas/MyCourses ahead of time.

Other Materials:

- A lab coat is required for lab. You may purchase your own or we *may* have some used ones available.

Classroom Policies

Evaluation:

Your overall course grade will be determined by calculating the % of points earned out of the possible 100. This will go towards the 600 total points for BIOL 414 as a whole. (The lab makes up 1/6 the total points for the course.)

Your overall course grade will be determined by calculating the percentage of points earned out of the possible 600. **The grade scale is:** 94-100%=A; 90-93%=A-; 87-89%=B+; 84-86%=B; 80-83%=B-; 76-79%=C+; 73-75%=C; 70-72%=C-; 66-69%=D+; 63-65%=D; 60-62%=D; Below 59.99%=F.

Extra credit policy:

In consultation with Dr. Cooke we may choose to offer a small extra credit assignment or two and these will be optional and offered to both lab sections. I will not give extra credit assignments to individual students to offset missing or incomplete assignments.

General Course Policies

Credit Hour Policy

This syllabus reflects the federal definition of a credit hour, which entails a minimum of 3 hours of engaged time per week per credit over a 15-week semester. Examples of engaged time include class time, assignments, examinations, laboratories, participation in course-related experiences (attending a talk or performance, speakers and events, fieldwork, etc.), conferences, and office hours. Student work reflects intended learning outcomes and is verified through evidence of student achievement. For more information, please see: [NECHE policy on Credits and Degrees](#).

Curtailed Operations

If the university curtails operations due to weather, we will not hold in-person activities for our safety and the safety of others. As soon as possible, I will post an announcement on Canvas about due dates, any make-up work, and (*if applicable*) any online options that may make sense depending on where we are in the course. Please make sure you have access to the UNH Alert RAVE system. If needed, sign up for RAVE Alerts [here](#).

Academic Alerts

The University is invested in your academic success. If a faculty member is concerned about your academic behavior or performance, they may submit an academic alert. Academic alerts are not punitive. The goal is to provide you with support and resources to support your success. They act as an important check-in point and, if you receive an academic alert, you will receive an email to your UNH email address. It is strongly recommended that you meet with a professional advisor and connect with your instructor to discuss the reason for the alert.

Academic Honesty

The Biology/Biotechnology Programs at UNH Manchester will strictly adhere to the [Academic Integrity Policy](#), as published in the UNH Student Rights, Rules, and Responsibilities Handbook. It is imperative that you review this policy to understand the rules and responsibilities about honesty, academic integrity, plagiarism, and cheating. By turning in any piece of work in this course, you declare that you have read and understand the policy, and that you did not engage in any form of academic dishonesty as defined in the Handbook.

Plagiarism can take many forms, such as submitting someone else's work - in whole or in part - as your own; collaborating on answers for individual homework assignments or allowing your own work to be used by another student; copying information from a web site or other text without proper documentation; buying a pre-written paper or lab report. *Your assignments submitted through Canvas will be routinely screened for plagiarism and AI use using sophisticated software.* It is strongly recommended that you complete the [Tutorial on Plagiarism in order to fully understand the definition\(s\) and all forms of it](#).

Cheating is mainly concerned with copying on quizzes, exams or in lab, bringing crib notes into an exam or referring to notes or the textbook or any other source such as a programmable calculator, tablet, or cell phone during an exam. All electronic devices must be turned off and placed upside down on the table in front of you for the duration of the exam. Ear buds are not permitted.

For online exams, you will be given instructions about which materials are and are not permitted to be used. You will be asked to digitally sign an Academic Honesty statement before beginning any exam agreeing to follow all exam policies.

Posting any portion of faculty-created materials such as homework assignments or exam questions to "tutoring" or "study" websites is a serious violation of the Academic Honesty policy of the University. Any instances of this form of cheating will be brought directly to the department chair and dean's office.

Any instances of cheating or plagiarism will result in consequences that can range from a failing grade on the assignment for all students involved to dismissal from the University, as defined in the UNH Student Rights, Rules, and Responsibilities Handbook.

AI Tools

Unless otherwise specified, the use of Automated Writing Tools, including Grammarly, ChatGPT and similar artificial intelligence (AI) tools, is strictly prohibited in this course, even when properly attributed. The use of automated writing tools to complete assignments is considered plagiarism (as defined by UNH's [Academic Integrity Policy](#)) and will be handled in accordance with existing policy.

To avoid concerns or misunderstandings, you're encouraged to:

- Complete all writing assignments using your own ideas and words.
- Turn on version history in Microsoft Word to demonstrate your iterative writing process.
- Ask if you're unsure whether a tool or method is acceptable for a given assignment.

In addition, other uses of AI are prohibited including:

- Using AI to transcribe course lectures
- Uploading course materials to an AI tool for the purposes of summary generation.

Attendance

Class attendance is important for your learning. Attendance in this lab course is mandatory. You are responsible for all course assignments and meeting all deadlines unless exceptions are agreed upon with the instructor ahead of time. If you need to miss class for a planned activity, let the instructor know ahead of time by email. Attendance will be noted during every lab section. If you must miss a lab for a valid reason (e.g., illness), please contact me (by email in advance) to let me know. Because participation is graded in this course, excessive absences will affect your grade and multiple unexcused absences will result in failure of the lab. Also, be sure to check MyCourses/Canvas for relevant Power Points, handouts, assignments, etc. See the [Attendance and Class Requirements policy](#) in the SRRR.

If a student needs accommodation for a religious or cultural holiday/observance, that student is encouraged to make that request as early in the semester as possible.

Temporary Academic Supports for Extended Absences with Letter

Suppose you're dealing with an unexpected, extenuating circumstance that will keep you out of class or affect your class performance for more than two days. In that case, you can email the [Dean of Students](#) (or [Stephanie Kirylych](#), Interim Assistant Dean of Student Success, for students at UNNH Manchester) to request that a Dean's Letter be sent to your faculty. In the email, include a brief description of the situation and provide some documentation (e.g., medical documentation). The Dean's Letter explains that you are experiencing compelling, non-academic circumstances beyond your control which might affect your academics. We ask that you reach out to your faculty to discuss how to stay on top of the work that you will need to miss.

When the letter is sent to your faculty, you will also receive a copy in your UNH email. An example of the letter can be found [here](#).

Please note that students may miss class or have an issue in class for a variety of reasons, and a notice does not need to be sent every time you need to miss one or two classes. Faculty have discretion to work with students who are managing personal issues. If you miss a day of class or are managing a situation that will affect your performance for that day, contact your faculty directly and let them know.

If you have questions about what is considered an excused absence or other attendance policy-related questions, please visit the [UNH Attendance Policy](#) page found in the Academic Catalog.

If you are required to miss significant class time, you will be provided with temporary academic supports so that you can continue to make satisfactory progress in this course. Please contact your course instructor to discuss the specific types of support that will be implemented during your absence, including:

- Class notes from a peer
- One-on-one meetings during office hours with instructor or TA to catch-up on missed content
- Lecture slides
- Handouts or other materials that are distributed

Classroom Conduct

Respect for Faculty, Learning Assistant(s) and Peers: During lab when faculty, assistants, or individual peers are talking, it is important that everyone else remain silent and listen. Questions posed by others in class are important, and mini presentations given by student groups are for the benefit of the *entire lab*, not only for the professor.

Cell phones: Please silence your cell phone before you enter the classroom.

Computer Use: You will sometimes want your laptop or tablet in the lab for relevant work. Be sure that you are using it for academic purposes and not other reasons.

Class Recording Policy

Lab lectures and activities will/may be recorded to make this course more accessible, and recordings may be made available to students presently enrolled in the class to assist in reviewing material and preparing for assessments. Recordings are to be used for educational use/purposes only. Students may **not** record class activities (including audio/photographs/screenshots of instructors and students) without express permission from their instructor and students whose image is captured. If permission is granted, student-initiated recordings may only be used for educational purposes in this class and only during the period in which the student is enrolled in the class; authorized student-initiated recordings may **not** be posted or shared in any fashion outside of the class.

COURSE/CAMPUS & LEARNING RESOURCES

Student Resources

UNH provides a variety of resources to support students' academic success, health and well-being, and engagement with the campus community. A list of student resources can be found on [the Provost's website](#). These resources are not required for this course but may be helpful to you during the semester.

Peer Tutoring Course Support

Knack is a peer-to-peer tutoring platform that is available to all enrolled students for all undergraduate courses in Durham and Manchester at no cost to students. Students looking for additional assistance outside of the classroom are advised to consider working with a peer tutor through Knack. UNH has partnered with Knack to provide students with access to verified tutors who have successfully completed your course. To view available tutors, visit unh.joinknack.com and sign in with your student account.

Connors Writing Center and Online Writing Lab

Connors Writing Center (CWC)

The [Connors Writing Center](#) offers one-on-one writing conferences to current UNH students (undergraduate and graduate). We work with writers from all disciplines on many kinds of writing. Our conferences are conducted by trained writing assistants who are UNH undergraduate and graduate students. The CWC offers both in-person and virtual appointments, available to any current UNH student. The online sessions are synchronous (real-time) appointments designed to help students have one-on-one conversations about their writing at any stage of the process, mirroring what we do during in-person appointments. [Students can schedule an appointment any time during the semester.](#)

Online Writing Lab (OWL)

The [Online Writing Lab \(OWL\)](#) assists students via asynchronous remote collaboration with qualified writing assistants. The OWL is not an editing or manuscript preparation service. Feedback may include reference to mechanics on the submitted paper, but it will be targeted to bring attention to patterns and issues to help the writer self-correct the paper and carry forward lessons to improve future writing. Students in all fields of study can submit up to 20 double-spaced pages of any form of writing, including course papers, resumes, creative essays, or personal statements. Within 3 business days, writing assistants provide feedback that can be printed or viewed on screen. The OWL is accessible through any computer that has an Internet connection and Microsoft Word.

To submit a paper, students can sign in at unh.mywconline.com and select the "Online Writing Lab" schedule.

Library Resources

The UNH Manchester librarians are available to assist you with the research process. Visit the library's website at <https://cps.unh.edu/library> to learn about library services and to search for reliable academic sources. You can contact the library at 603-641-4173 or at unhm.library@unh.edu.

The links below guide you to useful online library resources:

- Make a Research Appointment with a librarian: <https://libraryguides.unh.edu/remotearchive/researchhelp>
- Use the Library Search Box to find information: https://libraryguides.unh.edu/librarysearchbox_unhmanchester
- Reserve a Study Room: <https://cps.unh.edu/library/support-services>
- Discover resources for Citing Sources: <https://libraryguides.unh.edu/unhmcitingsources>
- Learn strategies for Evaluating Sources:
<https://libraryguides.unh.edu/ENGL401UNHManchester/evaluatingresources>.

Accessibility Services

According to the Americans with Disabilities Act (as amended, 2008), each student with a disability has the right to request services from UNH to accommodate his/her/their disability. If you are a student with a documented disability or believe you may have a disability that requires accommodations, please contact Student Accessibility Services (SAS) located on the Manchester campus in room 417 or sas.office@unh.edu.

Accommodation letters are created by SAS with the student. Please follow-up with your instructor as soon as possible to ensure timely implementation of the identified accommodations in the letter. Faculty have an obligation to respond once they receive official notice of accommodations from SAS but are under no obligation to provide retroactive accommodations.

For more information refer to www.unh.edu/sas or contact SAS at 603.862.2607, 711 (Relay NH) or sas.office@unh.edu.

Emotional or mental health distress

In partnership with The Mental Health Center of Greater Manchester, UNH Manchester offers consultation visits on a walk-in basis and through telehealth appointments.

Services include:

- Free confidential screening & consultation with a licensed mental health therapist.
- Referrals to mental health or substance misuse treatment. And assistance in understanding how to afford additional treatment (with or without insurance!) or find free services.

You may email: unhm.wellness@unh.edu to make an appointment to meet with a counselor by clicking [here](#)

If you would like to connect to counseling services directly, you may do so by contacting The Greater Manchester Mental Health Center at (603) 668 - 4111.

Confidentiality and Mandatory Reporting of Sexual Violence or Harassment

The University of New Hampshire and its faculty are committed to assuring a safe and productive educational environment for all students and for the university as a whole. To this end, the university requires faculty members to report to the university's [Title IX Coordinator](#) (Bo Zaryckyj, Bo.Zaryckyj@unh.edu, 603-862-2930/1527 TTY). Faculty, staff or students on the Manchester campus can also contact [Stephanie Kirylych](#), Interim Assistant Dean of Student Success and Deputy Title IX Coordinator (603) 641-4170 to report any incidents of sexual violence and harassment shared by students.

If you wish to speak to a confidential support service provider who does not have this reporting responsibility because their discussions with clients are subject to legal privilege, you can contact the [SHARPP Center for Interpersonal Violence Awareness, Prevention, and Advocacy](#) at (603) 862-7233/TTY (800) 735-2964, as well as, Caroline Young, SHARRP Center Advocacy Expanded Services Coordinator for UNH Manchester (caroline.young1@unh.edu; room 417; Available in

person Mondays 9 am to 4-pm and available by appointment (in person and virtually) by emailing caroline.young1@unh.edu). Individuals can also access Reach Crisis Services NH 603-668-2299 (24 hours), 77 Sundial Ave., Suite 306W, Manchester, NH.

For more information about what happens when you report, how the university treats your information once a report is made to the Title IX Coordinator, your rights and reporting options at UNH (including anonymous reporting options) please visit [student reporting options](#). [The uSafeUS app](#) is also available for students to keep reporting options and resources easily accessible on their phones.

Help us improve our campus and community climate. If you have observed or experienced an incident of bias, discrimination or harassment, please report the incident by contacting the Civil Rights & Equity Office at UNH.civilrights@unh.edu or TEL # (603) 862-2930 voice/ (603) 862-1527 TTY / 7-1-1 Relay NH, or visit the CREO website. Anonymous reports may be submitted.

Confidential Support Resources for UNH Manchester:

- SHARPP Extended Services Coordinator for UNH Manchester- room 417. Available in person Mondays 9 am to 4-pm and by Zoom appointment year-round M-F.
- YWCA, NH – 603-668-2299(24hour), 72 Concord St. Manchester, NH
- The Mental Health Center of Greater Manchester: On campus mental health counselors are available by appointment during the academic year. Located in room 426.
- 24 Hour NH Sexual Violence Hotline: 1-800-277-5570
- 24 Hour NH Domestic Violence Hotline: 1-866-644-3574

Basic Needs Resources:

UNH has many resources that can help students who are experiencing food insecurity, housing or other financial concerns. All enrolled students are eligible for Basic Needs resources and services. A complete listing of university and local basic needs resources can be found on the [UNH Manchester Basic Needs webpage](#). The following UNH resources students are referred to most frequently are listed below:

- **The Student Emergency Financial Assistance Fund (SEFA)** provides UNH undergraduate and graduate students unable to meet essential expenses due to a temporary or unexpected hardship with short-term financial assistance once per academic year. Please note that funding is currently limited due to high demand and UNH had to **pause funding for graduate students** at this time. We encourage graduate students to schedule a time to meet with the Basic Needs team to discuss their options and connect them to additional resources at basic.needs@unh.edu. UNH received a grant from Granite Edvance to support our Student Emergency Financial Assistance program specifically for undergraduate students. To request SEFA students must complete the [Student Emergency Financial Assistance Request Form](#)
- **UNH Manchester Wildcat Cupboard**, located on the second floor, Room 290, is available to all UNH Manchester undergraduate and graduate students, staff or faculty who need access to free, nourishing food. There are an assortment of items including pasta, canned food, cereal, milk, eggs, fresh fruits and vegetables, which are purchased from the NH Food Bank through a grant or donated by members of the community. The Cupboard is open Tuesdays and Thursdays from 12:30-3:30 pm and Wednesdays from 3:00-6:00 pm. Walk ins are welcomed and no appointments are needed.
- **The UNH Manchester Wildcat Wardrobe** provides students with access to new and gently used professional attire for interviews, career fairs, networking events, and the workplace. Students can visit the Wildcat Wardrobe, located on the second floor in Room 288 (next to the Wildcat Cupboard) Monday-Friday from 9-5 pm. Items are completely free.

LAB ASSIGNMENTS AND POINTS

Of the 100 possible lab points, the points will be earned through these assignments and activities:

- **Lab worksheets 50%**
- **Experimental/lab summary assignments 20%**
- **Final presentation or poster 15%**
- **Paper lab notebook 10%**
- **Group participation evaluation 5%**

Tentative Lab Schedule – Spring 2026

Starting 1/27, labs will meet 9:10am (except when noted) in Room 380 for a starting presentation. We will then head to lab in 554 for lab activities.

Date	Topic
1/20	Review Lab Schedule and Syllabus; Safety; Overview of Diversity of Life
1/27	Domains Bacteria & Archaea; Bacteria and Early Life Part 1
2/3	Kingdom Protista; Bacteria Part 2
2/10	Kingdom Fungi; Unicellular Eukaryotic Life
2/17	Kingdom Plantae; Plants Part 1
2/24	NO LAB MEETING
3/3	Kingdom Plantae (cont'd); Plants Part 2
3/10	Plants Part 3
3/17	Spring Break—NO LAB MEETING
3/24	Kingdom Animalia; Animals Part 1
3/31	Kingdom Animalia (cont'd); Animals Part 2
4/7	Kingdom Animalia (cont'd); Animals Part 3
4/14	Kingdom Animalia (cont'd); Animals Part 4 and Ecology Part 1
4/21	Kingdom Animalia (cont'd); Animals Part 5 and Ecology Part 2
4/28	Final lab presentations